



MATTEO CAVICCHIOLI

BIOMEDICAL & AI ENGINEER

AI DEVELOPER · TRANSLATIONAL THINKING ·
CREATIVITY · COMPUTER VISION ·
INNOVATION · REAL-WORLD IMPACT

INFO



CASSANO MAGNAGO, VARESE (IT)



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• PROFESSIONAL SUMMARY •

Biomedical engineer with a strong foundation in AI, electronics, and physiological signals. PhD candidate at Politecnico di Milano, working on AI advanced 3D anatomical reconstruction for clinical use. Played a key role in research and technology development that led to the creation and initial funding of a med-tech startup. Experienced in cross-disciplinary collaboration with clinicians, engineers, and business stakeholders, with a hands-on approach to translating research into real-world healthcare impact.

• EDUCATION •

PH.D. IN BIOMEDICAL ENGINEERING

POLITECNICO DI MILANO | 2022 – **DEC 2025**

Focus: AI for Digital Twins Reconstruction

M.SC. BIOMEDICAL ENGINEERING

POLITECNICO DI MILANO | 2019 – 2022

Spec.: Technologies for Electronics (BTE)

B.SC. BIOMEDICAL ENGINEERING

POLITECNICO DI MILANO | 2015 – 2019

• TECHNICAL SKILLS •

Programming: Python, MATLAB, C, SQL

AI: TensorFlow, PyTorch, MONAI

Development Tools: Git, Bitbucket

DevOps & Cloud: Docker, AWS, Aruba Cloud

Hardware: Arduino, Raspberry Pi

• EXPERIENCE •

AI ENGINEER – AIMS ACADEMY FOUNDATION

MILAN, ITALY | 2022 – **PRESENT**

- Developed AI-driven solutions & computer vision models for surgical planning.
- Developed a D²-RD-UNet model for multiclass liver vessel segmentation, improving 3D reconstruction accuracy.
- Automatic vessel centerline extraction, radius branching, and tubular rendering.
- IT infrastructure manager.

RESEARCH INTERN – UCSF & POLIMI

SAN FRANCISCO (US) & MILAN | 2021 – 2022

- Worked on dynamic speech MRI data to classify neurodegenerative diseases affecting vocal tract.

PIZZA CHEF HELPER

ITALY | 2015 – 2022

- Developed strong work ethic, time management, and collaborative skills

• PUBLICATIONS •

- "Main Challenges on the Curation of Large Scale Datasets for Pancreas Segmentation Using Deep Learning in Multi-Phase CT Scans: Focus on Cardinality, Manual Refinement, and Annotation Quality." **Computerized Medical Imaging and Graphics** 117 (2024).
- "Deep Learning for Pancreas Segmentation on Computed Tomography: A Systematic Review." **Artificial Intelligence Review** 58.8 (2025)
- "D²-RD-UNet: A dual-stage dual-class framework with connectivity correction for hepatic vessels segmentation." **Computers in Biology and Medicine** (2025).